Historical USDA records

taxon, region, region, ...

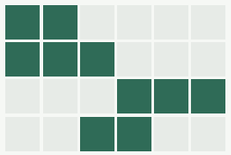
recorded plant presence

US states + Canadian provinces

Normalize rank & region codes

- Remove genus summary rows
- Merge infraspecific records
- Correct codes; merge NF + LB

Species × 64-region matrix



1 recorded

0 unrecorded

Freeze discovery / validation

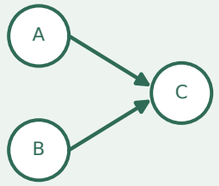
80% training **20% hold-out**

Training species

Region association mining

FP-growth | itemset length ≤ 3

Filter by support, confidence and lift

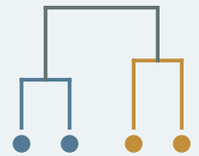


Single-region and two-region antecedents

Regional composition clustering

Jaccard distance | average linkage

Choose k by silhouette (k = 4–9)



Groups of regions with shared recorded flora

Reliability checks and extension experiments

Replication & record completion

Random / genus splits; hide one recorded region

Bootstrap stability

Rule recurrence + cluster co-assignment

Broad-distribution sensitivity

Remove the broadest 5% of species

Algorithm agreement & efficiency

Apriori vs FP-growth: same itemsets

Distance / linkage sensitivity

Jaccard vs cosine; average vs complete

Spatial permutation test

Composition distance vs geographic distance